

CS 2026 ACADEMIC PROPOSALS

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INTRODUCTION

This document provides a template for submitting *Session Proposals* and *Tutorial Proposals* to the Chiplet Summit 2026. It is designed to be shared with academic colleagues at Cornell University, UC Berkeley, Stanford University, Georgia Tech and the University of Illinois Urbana–Champaign (UIUC). The structure highlights both the practical engineering challenges of chiplets and the academic advances that provide rigor, vision, and excitement.

SESSION PROPOSAL

TITLE

Chiplet Design and Integration: Academic Advances in Placement, Interconnect Modeling, and Reconfigurability

ABSTRACT

This session brings together leading academic voices to present breakthroughs in chiplet system design. Topics include chiplet placement optimization, interconnect modeling at terabit scales, reconfigurable architectures for heterogeneous workloads, and design space exploration methods. The session aims to balance near-term practitioner challenges (verification, interoperability, packaging constraints) with long-term research trajectories that could shape the chiplet ecosystem over the next decade.

TARGET AUDIENCE

Industry engineers seeking state-of-the-art design methodologies, academic researchers, and students interested in bridging theory and practice.

PROPOSED SPEAKERS

- **Prof. Hsien-Hsin (Sean) Lee, Cornell University.** Research on chiplet interconnect modeling and scalable design exploration. [Google Scholar Profile](#).
- **Prof. Krste Asanović, UC Berkeley.** Co-creator of RISC-V; work on open architectures and chiplet-based compute fabrics. [Google Scholar](#)

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[Profile.](#)

- **Prof. Mark Horowitz, Stanford University.** Pioneer in VLSI and interconnect scaling; leadership in chiplet-based memory–logic integration. [Google Scholar Profile.](#)
- **Prof. Rakesh Kumar, UIUC.** Research on reconfigurable microarchitectures and heterogeneous integration. [Google Scholar Profile.](#)

OUTCOMES

1. Present a rigorous survey of academic advances relevant to chiplet practitioners.
2. Highlight opportunities for industry–university collaboration.
3. Inspire next-generation researchers through exposure to industrially relevant challenges.

TUTORIAL PROPOSAL TEMPLATE

TITLE

Chiplet Systems: From Theory to Practice

ABSTRACT

This tutorial introduces chiplet system design from both industrial and academic perspectives. Attendees will gain practical skills in chiplet placement strategies, interconnect modeling tools, and design space exploration frameworks. In addition, the tutorial highlights reconfigurability techniques, verification workflows, and emerging trends such as 3D chiplets and photonic interconnects.

TARGET AUDIENCE

Practicing engineers, graduate students, and researchers interested in hands-on methodologies for chiplet integration.

PROPOSED INSTRUCTORS

- Faculty from Cornell, UC Berkeley, Stanford, and UIUC (names as above).
- Graduate students/postdocs to lead tool demonstrations (design space exploration, modeling frameworks).

LEARNING OBJECTIVES

- Understand academic approaches to chiplet placement and interconnect modeling.
- Apply design space exploration tools to evaluate cost/power/latency tradeoffs.
- Explore reconfigurable chiplet architectures for emerging AI and HPC workloads.

REFERENCES

- Asanović, K. et al., *The RISC-V Instruction Set Manual*, UC Berkeley.
- Horowitz, M. et al., seminal work on VLSI systems and interconnect scaling.
- Selected Google Scholar profiles for invited speakers.

CONCLUSION

This proposal template is intended as a flexible framework for engaging top academic contributors. Each university can adapt the session and tutorial scope to emphasize their unique contributions while aligning with the Chiplet Summit's dual mandate: solve practitioner problems and explore bold, forward-looking research.